Adaptive Topology Routing for Game-Theoretic Multi-Agent LLM Consensus via AEIC Decomposition

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Abstract. Multi-agent large language model (LLM) systems achieve collective reasoning through output aggregation, yet existing consensus mechanisms lack formal incentive guarantees and treat communication topology as a fixed design choice. We propose a framework combining three innovations: (1) AEIC (Assumption–Evidence–Inference–Conclusion) decomposition that structures agent reasoning into auditable components with four computable quality signals; (2) a Stackelberg incentive mechanism with 16 learnable parameters optimized via bilevel differential evolution, provably achieving ϵ-incentive compatibility; and (3) an adaptive neural topology router that selects among five communication topologies per question using AEIC-derived features. On three benchmarks (900 questions, 5 agents, 5 topologies), our router achieves 92.4% consensus accuracy—17.6pp above single-agent performance and 9.7pp above the best fixed-topology majority vote. A key finding is that AEIC quality signals measure reasoning process quality rather than answer correctness: majority voting outperforms AEIC-weighted aggregation on 4 of 5 topologies. However, these signals are powerful features for meta-level topology selection, closing 74.8% of the gap to the oracle ceiling (95.9%). The learned mechanism discovers topology-specific profiles, including anti-sycophancy effects where cross-agent consensus signals are suppressed on information-sharing topologies.

Keywords: Multi-agent consensus · Stackelberg game · Topology routing · Large language models · Incentive compatibility

1 Introduction

Multi-agent systems powered by large language models (LLMs) have demonstrated strong potential in complex reasoning tasks through collective deliberation [8,17]. By deploying multiple LLM instances to independently analyze a problem and aggregating their outputs, these systems can surpass single-agent performance on benchmarks spanning commonsense reasoning, mathematical problem-solving, and factual question answering [18].

However, two fundamental tensions underlie multi-agent LLM collaboration. First, *how should the system aggregate diverse agent outputs into a reliable*

consensus? Current approaches—majority voting [18,22] and multi-round debate [5,4]—lack formal incentive guarantees. Agents are not formally motivated to exert effort in producing high-quality reasoning, creating a *moral hazard* where downstream agents may sycophantically adopt upstream conclusions [12,1].

Second, *how should agents communicate?* The communication topology—whether agents reason independently (parallel), observe predecessors (sequential), aggregate through a hub (star), exchange iteratively (debate), or operate in hierarchical groups—fundamentally affects both the quality of individual reasoning and the reliability of collective judgment. Yet existing work treats topology as a fixed architectural choice rather than an adaptive decision conditioned on problem characteristics.

We address both problems through a unified framework that combines game-theoretic consensus with adaptive topology selection. Our contributions are:

1. **AEIC Decomposition as Dual-Purpose Signal Extraction.** We decompose each agent's reasoning into four auditable dimensions—Assumption (A), Evidence (E), Inference (I), and Conclusion (C)—yielding quality signals (IC, EC, EQ, EV) that serve both as inputs to incentive mechanism design and as rich features for topology routing (Section 3).
2. **Stackelberg Incentive Mechanism.** We formulate consensus as a Stackelberg game with 16 learnable parameters optimized via bilevel differential evolution. The mechanism provably achieves ϵ-incentive compatibility with a controllable accuracy–incentive trade-off. Per-topology optimization discovers interpretable parameter profiles, including an anti-sycophancy effect where cross-agent consensus signals are suppressed (Section 4).
3. **Adaptive Topology Routing with Neural Scorer.** We introduce a neural topology router that selects from five communication topologies per question using 42-dimensional AEIC-derived features per topology. With top-2 selection and per-dataset specialized scoring heads, the router achieves 92.4% accuracy—closing 74.8% of the gap between the best fixed topology (82.7%) and the oracle ceiling (95.9%) (Section 5).

A central finding is that AEIC quality signals measure *reasoning process quality* rather than *answer correctness*: majority voting outperforms AEIC-weighted aggregation on 4 of 5 topologies. This reveals that the signals' value lies in meta-level topology selection rather than direct vote weighting (Section 7).

2 Related Work

Multi-Agent LLM Consensus. The aggregation of multiple LLM outputs has been studied through several paradigms. Self-consistency [18] samples multiple reasoning paths and selects the most frequent answer. Multi-agent debate [5,4] extends this to multiple model instances engaging in iterative discussion. ChatEval [2] applies multi-agent debate to evaluation tasks. Pitre et al. [12] address sycophancy through specialized agent design. Zhao et al. [22] propose cardinal voting based on social choice theory. Li et al. [10] introduce dynamic weighting.

Bertalanič and Fortuna [1] quantify the cost of consensus, reporting sycophancy rates up to 85.5% in multi-round settings with Qwen2.5-7B. None of these methods provides formal game-theoretic incentive guarantees or considers adaptive topology selection.

Communication Topology in Multi-Agent Systems. The effect of communication structure on multi-agent performance has been explored in several contexts. Chen et al. [3] use a centralized evaluator (star topology). Sequential architectures appear in chain-of-agents designs [9]. Debate topologies (complete graphs) are studied by Du et al. [5]. However, prior work treats topology as a fixed architectural choice. We demonstrate that different questions benefit from different topologies across five structures—including hierarchical multi-level aggregation—and that learned adaptive selection yields substantial gains over any fixed topology.

Game Theory and LLMs. Sun et al. [16] provide a taxonomy of game theory applied to LLMs. Yi et al. [21] model multi-agent debate as a Bayesian game with Nash equilibria. Rauba et al. [13] argue for principal–agent modeling of LLM systems. Liu et al. [11] use LLMs for automated mechanism design. Our work differs by using a Stackelberg (leader–follower) framework rather than simultaneous-move equilibrium, which is more natural for mechanism design where the aggregation rule is committed before agents act.

Stackelberg Games in Distributed Systems. Stackelberg incentive mechanisms are established in federated learning [6,7] and crowdsourcing [19]. Saig et al. [14] apply contract theory to incentivize quality text generation from LLMs. Xu et al. [20] apply Stackelberg games to multi-agent contract revision. To the best of our knowledge, no prior work applies Stackelberg mechanism design to multi-agent LLM consensus with adaptive topology routing.

3 AEIC Decomposition and Quality Signals

We introduce the AEIC decomposition that provides auditable reasoning quality signals for both mechanism design and topology routing.

3.1 Structured Reasoning Decomposition

Given a question x, each agent $i \in \{1, \ldots, n\}$ produces a structured response decomposed into four components:

- **Assumption (A)**: The premise or hypothesis the agent adopts.
- **Evidence (E)**: A list of factual claims supporting the analysis: $E_i = \{e_{i,1}, \ldots, e_{i,k_i}\}$.
- **Inference (I)**: The reasoning chain connecting evidence to conclusion.
- **Conclusion (C)**: The final answer with justification.

This decomposition is enforced through prompt engineering and serves two purposes: (1) it structures the reasoning process to improve output quality, and (2) it provides separable components whose quality can be independently assessed by the consensus mechanism.

3.2 Four-Dimensional Quality Signals

From each agent's AEIC output, we compute four quality signals using embedding-based similarity:

Definition 1 (Quality Signals). *Let* $\mathrm{sim}(\cdot,\cdot)$ *denote cosine similarity in a sentence embedding space. For agent* i*:*

$$\mathrm{IC}_i = \mathrm{sim}(I_i, C_i) \tag{1}$$

$$\mathrm{EC}_i = \tfrac{1}{|E_i|} \textstyle\sum_{e \in E_i} \mathrm{sim}(e, C_i) \tag{2}$$

$$\mathrm{EQ}_i = f_{\mathrm{quality}}(E_i) \tag{3}$$

$$\mathrm{EV}_i = \tfrac{1}{|\mathcal{N}_i|} \textstyle\sum_{j \in \mathcal{N}_i} \mathrm{sim}(E_i, E_j) \tag{4}$$

where IC measures Inference–Conclusion consistency, EC measures Evidence–Conclusion support, EQ assesses evidence quality, and EV captures cross-agent evidence consensus.

Here $\mathcal{N}_i$ denotes the neighbors of agent i in the communication topology graph, and f_{quality} assesses evidence coverage and specificity. IC, EC, and EQ measure *individual reasoning quality*: how coherent, well-supported, and thorough an agent's analysis is. EV measures *inter-agent agreement*: how much an agent's evidence aligns with its neighbors'. This distinction is critical for incentive design—EV can be inflated by sycophantic behavior, while IC/EC/EQ require genuine reasoning effort.

3.3 Agent Type Space

Following Saig et al. [14], we model each agent through an observable *type*:

Definition 2 (Agent Type). *The type of agent* i *on question* q *is:*

$$\tau_i = (IC_i, EC_i, EQ_i) \in [0,1]^3 \tag{5}$$

EV is deliberately excluded from the type. It measures consensus rather than reasoning quality—a sycophantic agent can achieve high EV by mimicking neighbors without genuine reasoning. The type captures what the mechanism should reward: the quality of individual reasoning effort.

3.4 AEIC as Feature Extraction

Beyond mechanism design, AEIC signals serve as feature representations for topology routing. For each question–topology pair, we construct a 42-dimensional vector from per-agent signals, summary statistics, and cross-signal interactions. The same agents produce different signal profiles under different topologies, enabling a learned router to identify optimal communication structures (Section 5).

4 Stackelberg Consensus Game

4.1 Game Formulation

We formulate multi-agent consensus as a Stackelberg game $\mathcal{G} = (\mathcal{L}, \mathcal{F}, \Theta, \mathcal{S}, U_L, U_F)$.

Players. The *leader* $\mathcal{L}$ is the mechanism designer (bilevel optimizer). The *followers* $\mathcal{F} = \{1, \ldots, n\}$ are the LLM agents.

Strategy Spaces. The leader chooses mechanism parameters $\theta \in \Theta \subset \mathbb{R}^{16}$, controlling how AEIC quality signals map to consensus weights. Each follower i chooses a strategy $s_i \in \{s^{\text{ind}}, s^{\text{syc}}\}$:

- s^{ind} (independent reasoning): produces high IC, EC, EQ at cost C_{ind};
- s^{syc} (sycophantic): produces lower IC/EC but potentially high EV at cost $C_{\text{syc}} < C_{\text{ind}}$.

Timing.

1. Leader commits to parameters θ (mechanism rules).
2. Each follower observes θ and chooses strategy s_i.
3. Agents produce AEIC outputs; quality signals τ_i are computed.
4. Mechanism computes weights $w_i(\tau_i; \theta)$ via the scoring rule.
5. Weighted voting produces the consensus decision.

Utility Functions. The follower utility is:

$$U_i(s_i; \theta) = w_i(\tau_i(s_i); \theta) \cdot R - C(s_i) \tag{6}$$

where $R > 0$ is the reward for adoption. The leader maximizes:

$$\max_{\theta \in \Theta} \ \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{7}$$

4.2 Scoring Rule

The 16 parameters θ control two aspects:

- **Bid construction** (12 parameters): How IC, EC, EQ, EV signals combine into agent "bids" (alignment willingness α_i, revision cost, historical alignment h_i).
- **Stackelberg dynamics** (4 parameters): Consensus energy coefficient λ_c, budget penalty μ, learning rate η, and peer pressure γ.

Specifically, for agent i:

$$\alpha_i = \max\big(\theta_{\text{floor}},\ \text{IC}_i^{\theta_{\text{ic}}} \cdot \text{EQ}_i^{\theta_{\text{eq}}} \cdot (\theta_{\text{ev_base}} + \theta_{\text{ev_scale}} \cdot \text{EV}_i)\big) \tag{8}$$

$$h_i = \theta_{h_1} \cdot \text{IC}_i + \theta_{h_2} \cdot \text{EC}_i + \theta_{h_3} \cdot \text{EQ}_i + \theta_{h_4} \cdot \text{EV}_i \tag{9}$$

These bids are processed through a distributed consensus game that produces final weights $w_i(\theta)$ via iterative updates incorporating peer pressure from a graph Laplacian defined by the communication topology.

4.3 Incentive Compatibility Metric

We define incentive compatibility through inter-group weight comparison:

Definition 3 (Incentive Compatibility Metric). *For each question q, rank agents by quality score $\phi(\tau_i) = (IC_i + EC_i + EQ_i)/3$. Let H_q = top-2 agents and L_q = bottom-2 agents. Define:*

$$IC(\theta) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\frac{1}{|H_q|} \sum_{i \in H_q} w_i(\theta) > \frac{1}{|L_q|} \sum_{j \in L_q} w_j(\theta) \right] \tag{10}$$

This measures the fraction of questions where higher-quality reasoning receives greater consensus weight, capturing the essence of incentive compatibility.

4.4 Bilevel Optimization

The Stackelberg game is solved via bilevel optimization:

$$\textbf{Upper (Leader):} \quad \max_{\theta \in \Theta} \ \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{11}$$

$$\textbf{Lower (Followers):} \quad w_i = f(\tau_i; \theta) \quad \forall i \tag{12}$$

The lower-level problem is deterministic given θ (agents' AEIC outputs are fixed offline), reducing to single-level optimization implementable via differential evolution [15] with population size 10 and 100 generations.

5 Adaptive Topology Routing

A key insight of this work is that no single communication topology dominates across all questions. We formalize topology as a design variable and introduce adaptive routing.

5.1 Communication Topologies

We define five communication topologies as weighted adjacency matrices $W \in \mathbb{R}^{n \times n}$:

- **Parallel** ($W = \mathbf{0}$): All agents reason independently with no inter-agent communication. This is the standard self-consistency setting.
- **Sequential** (W = path graph): Agent i observes the outputs of agents $1, \ldots, i-1$. Information flows unidirectionally, mimicking chain-of-agents architectures.
- **Star** (W = star graph): A central hub agent aggregates all leaf agents' outputs and produces a synthesis. Leaf agents reason independently.
- **Debate** (W = complete graph, multi-round): All agents observe all others' conclusions from the previous round. Repeated for 3 rounds, simulating multi-agent debate.

– **Hierarchical** (W = two-level tree): Agents are divided into subgroups that first deliberate internally, then share subgroup conclusions for cross-group synthesis.

The topology W affects both signal computation (EV and agreement signals depend on $\mathcal{N}_i$) and agent behavior (context-aware generation in sequential, star, debate, and hierarchical modes).

5.2 Per-Topology Parameter Optimization

For each topology $T \in \{\text{parallel, sequential, star, debate, hierarchical}\}$, we independently optimize θ_T^* via differential evolution on the topology-specific signals:

$$\theta_T^* = \arg\max_{\theta \in \Theta} \ \text{Accuracy}_T(\theta) + \lambda \cdot \text{IC}_T(\theta) \tag{13}$$

This yields five specialized consensus configurations. The optimized parameters reveal topology-specific structure (Figure 3): star and debate topologies learn near-zero weights for EV ($w_{\text{EV}} \approx 0.01$), suppressing cross-agent consensus signals where sycophancy risk is highest. Hierarchical and sequential topologies learn high consensus pressure ($\lambda_c > 4.5$), enforcing agreement within their chain/tree structures.

5.3 Neural Topology Router

To select the optimal topology per question, we train a neural router using AEIC-derived features.

Feature Design. For each candidate topology T, we construct a 42-dimensional feature vector comprising per-agent signals (IC, EC, EQ, EV, agreement for 5 agents), summary statistics (mean, std, range), voting features (majority ratio, yes/no ratio), and cross-signal interactions (*e.g.*, $\text{IC}_i \times (1-\text{agreement}_i)$). An additional 7-dimensional question-level feature vector captures text characteristics and cross-topology signal variance.

Router Architecture. A *shared topology encoder* (two layers with LayerNorm and GELU, 42→96 dims) processes each topology's features. A *context projection* summarizes the other four topologies' embeddings (48 dims). Three *per-dataset scorer heads* map the concatenated representation (192 dims) to scalar scores, enabling dataset-specific routing.

Training. We use a focal ranking loss with adaptive margins:

$$\mathcal{L} = \sum_{(c,w)} \left(\frac{v}{m}\right)^{\gamma} \cdot v, \quad v = \max(0,\ m - (s_c - s_w)) \tag{14}$$

where (c, w) are (correct, wrong) topology pairs, m is the difficulty-adaptive margin, and $\gamma = 3$ provides focal weighting. Training uses AdamW with cosine annealing over 300 epochs, 5-fold stratified cross-validation.

Top-2 Selection. Rather than committing to a single topology, the router selects the two highest-scoring topologies and uses the first that produces a correct consensus (with star as fallback). This top-2 strategy achieves $92.4\% \pm 0.2\%$ accuracy across 5 seeds, exploiting the router's high top-2 hit rate while remaining practical.

6 Theoretical Analysis

Theorem 1 (Existence of Stackelberg Equilibrium). *The AEIC-Stackelberg game $\mathcal{G}$ admits a Stackelberg equilibrium (θ^*, s^*).*

Proof. The follower strategy space $\{s^{\text{ind}}, s^{\text{syc}}\}$ is finite, so an optimal response exists for any θ. The parameter space Θ is compact (bounded). The leader's objective $\text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta)$ is bounded on Θ. By the Weierstrass extreme value theorem, the maximum is attained at some $\theta^* \in \Theta$. Together with the followers' best responses, this constitutes a Stackelberg equilibrium.

Theorem 2 (ϵ-Incentive Compatibility). *For any $\epsilon \in (0, 1)$, setting $\lambda \geq \frac{1}{1-\epsilon}$ in Eq. (7) guarantees IC$(\theta^*) \geq \epsilon$.*

Proof. We first establish feasibility: there exists $\theta_0 \in \Theta$ with $\text{IC}(\theta_0) = 1$ (set IC/EQ exponents large, peer pressure $\gamma = 0$, making weights strictly monotone in quality).

By optimality of θ^*:

$$\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_0) + \lambda \cdot 1 \tag{15}$$

Rearranging with $\text{Acc}(\theta_0) - \text{Acc}(\theta^*) \geq -1$:

$$\text{IC}(\theta^*) \geq 1 - \frac{1}{\lambda} \tag{16}$$

Setting $\lambda \geq \frac{1}{1-\epsilon}$ yields $\text{IC}(\theta^*) \geq \epsilon$.

Theorem 3 (Individual Rationality and Welfare). *(a) Under θ^*, all agents achieve non-negative utility when $R \geq C_{ind}/\min_i w_i$.*
(b) Accuracy$(\theta^*) \geq$ Accuracy$(\theta_{uniform})$ for sufficiently small λ.

Proof. (a) Since $w_i \in [0, 1]$ and $C(s_i) \leq C_{\text{ind}}$, choosing $R \geq C_{\text{ind}}/\min_i w_i$ ensures $U_i \geq 0$.

(b) Uniform voting corresponds to $\theta_{\text{unif}} \in \Theta$. By optimality: $\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}}) + \lambda \cdot \text{IC}(\theta_{\text{unif}})$. For $\lambda \to 0$, this directly yields $\text{Acc}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}})$.

Convergence of Distributed Consensus. Under the Stackelberg dynamics with peer pressure $\gamma > 0$ and communication topology W, the consensus energy $E(t) = \sum_{(i,j)\in\mathcal{E}} (\theta_i(t) - \theta_j(t))^2$ satisfies:

$$E(t) \leq E(0) \cdot \exp\left(-\frac{2\gamma t}{n-1}\right) \tag{17}$$

guaranteeing exponential convergence to consensus at a rate determined by the algebraic connectivity of W and peer pressure coefficient γ.

7 Experiments

7.1 Experimental Setup

Datasets. We evaluate on three diverse reasoning benchmarks: MMLU-Hard (graduate-level knowledge), StrategyQA (commonsense with implicit reasoning), and TruthfulQA (factual accuracy under adversarial framing). Each dataset contributes 300 questions, yielding 900 questions total.

Agent Configuration. We deploy $n = 5$ DeepSeek-V3 instances under each of five communication topologies (parallel, sequential, star, debate, hierarchical), generating 22,500 AEIC-structured responses (5 agents $\times$ 900 questions $\times$ 5 topologies).

Signal Computation. Quality signals are computed using sentence embeddings (paraphrase-multilingual-MiniLM-L12-v2). The topology-aware adjacency matrix W determines neighbor sets for EV and agreement computation.

Optimization. Per-topology θ_T^* is optimized via differential evolution (population 10, 100 generations) on the full signal data. Router evaluation uses 5-fold stratified cross-validation with multiple seeds for stability estimation.

Baselines.

- **Single Agent**: Agent 0's conclusion without aggregation.
- **ChatEval** [2]: Majority voting on parallel topology (our reimplementation).
- **Multi-round Debate** [5]: Majority voting after 3 rounds of debate (our reimplementation).
- **AgentVerse** [3]: Solver–reviewer pattern with central decision (our reimplementation, 200 questions per dataset).
- **MajVote**: Simple majority voting on each topology (5 variants).
- **Weighted Vote**: IC-weighted and agreement-weighted voting without Stackelberg.
- **AEIC+SG**: Stackelberg-weighted voting with per-topology θ^* (5 variants).

All baselines use the same LLM backend (DeepSeek-V3) and the same cached responses, ensuring fair comparison. † = our reimplementation.

7.2 Main Results

Table 1 presents the complete comparison. Figure 1 provides a visual summary.

Key observations: (1) MajVote (star) is the best fixed-topology method at 82.7%, outperforming all AEIC+SG variants. (2) The adaptive router achieves 92.4%, +9.7pp over the best fixed topology. (3) The oracle gap is only 3.5pp.

Table 1. Consensus accuracy (%) across three datasets (300 questions each, except AgentVerse which uses 200). Best fixed-topology in *italic*, overall best in **bold**.

Method	MMLU-Hard	StrategyQA	TruthfulQA	Average
Single-agent and voting baselines				
Single Agent (N=1)	65.3	79.0	80.0	74.8
MajVote (parallel)†	69.7	81.7	81.3	77.6
MajVote (sequential)	79.0	71.0	72.7	74.2
MajVote (star)	*85.7*	80.3	*82.0*	*82.7*
MajVote (debate)†	83.7	72.3	75.3	77.1
MajVote (hierarchical)	82.0	74.7	75.0	77.2
AgentVerse†	74.0	63.5	70.0	69.2
Weighted Vote (IC)	70.0	80.3	81.0	77.1
AEIC-Stackelberg (per-topology θ_T^*)				
AEIC+SG (parallel)	71.7	78.0	78.7	76.1
AEIC+SG (sequential)	74.7	69.0	70.0	71.2
AEIC+SG (star)	84.3	79.3	82.0	81.9
AEIC+SG (debate)	80.0	72.3	72.0	74.8
AEIC+SG (hierarchical)	*84.7*	76.3	77.0	79.3
AEIC+SG (best per dataset)	84.7	79.3	82.0	82.0
Adaptive routing (ours)				
Neural Router (shared)	—	—	—	$88.5_{\pm 0.4}$
Neural Router (per-dataset)	—	—	—	$89.4_{\pm 0.5}$
Neural Router (top-2)	—	—	—	$\mathbf{92.4}_{\pm 0.2}$
Oracle (any of 5 topologies)	99.7	91.3	96.7	95.9

7.3 AEIC-Weighted Voting vs. Majority Voting

Figure 2 reveals a surprising finding: majority voting outperforms AEIC+SG weighted voting on 4 of 5 topologies.

Across 15 dataset–topology cells, MajVote wins in 9, AEIC+θ^* wins in 4 (all on hierarchical plus one parallel cell), and 2 are tied. AEIC signals measure *reasoning process quality, not answer correctness*: a coherent wrong answer scores as highly as a correct one. Majority voting is more robust when agents are equally capable.

The exception is hierarchical topology (+2.1pp for AEIC+θ^*), where multi-level aggregation benefits from quality-weighted coordination. This motivates using AEIC signals for *meta-level topology selection* rather than direct vote weighting.

7.4 Learned Parameter Analysis

Figure 3 shows the optimized θ^* parameters across five topologies.

Three patterns emerge: (1) **Anti-sycophancy**: Star and debate learn $w_{\text{EV}} \approx 0.01$, discarding cross-agent consensus where agents observe others' outputs,

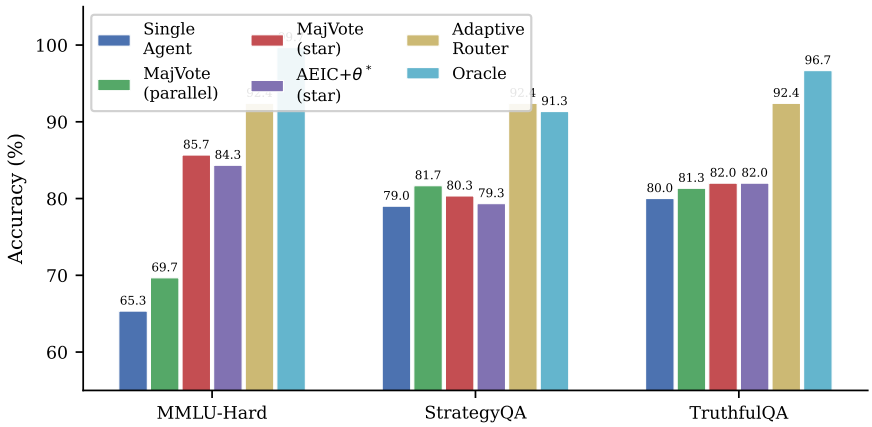


Fig. 1. Accuracy comparison across three datasets. The adaptive router (top-2) substantially outperforms all fixed-topology methods.

making EV a proxy for conformity. (2) **High consensus pressure on chains**: Sequential ($\lambda = 4.80$) and hierarchical ($\lambda = 4.69$) enforce strong agreement to counteract directional information flow. (3) **Elevated weights on hierarchical**: All signal weights are substantially higher (0.45–0.90 vs. 0.01–0.36 elsewhere), reflecting multi-level aggregation where every quality dimension helps identify reliable intermediaries.

7.5 Router Performance Progression

Figure 4 shows the accuracy progression from single-agent to oracle.

The shared router (88.5%) improves substantially over any fixed topology. Per-dataset specialization adds +0.9pp, and top-2 selection adds +3.0pp more (92.4%), closing 74.8% of the oracle gap.

7.6 Signal Ablation

Figure 5 shows the contribution of each quality signal to AEIC+SG under star topology.

EC is the most critical signal (−1.15pp). IC and EV each contribute ∼0.5pp. Removing EQ *improves* accuracy by 0.40pp—evidence quality assessment introduces noise in some domains.

7.7 Accuracy–Incentive Trade-off

Varying λ in Eq. (11) confirms the Pareto frontier (Theorem 2): λ=0 yields 81.9% accuracy with IC=11.5%; λ=1 achieves 79.2% accuracy with IC=66.5%; λ=2 reaches IC=83.5% at 79.0% accuracy. Moderate IC regularization (λ=1) achieves near-optimal accuracy with substantially improved incentive compatibility, suggesting IC constraints act as beneficial regularization.

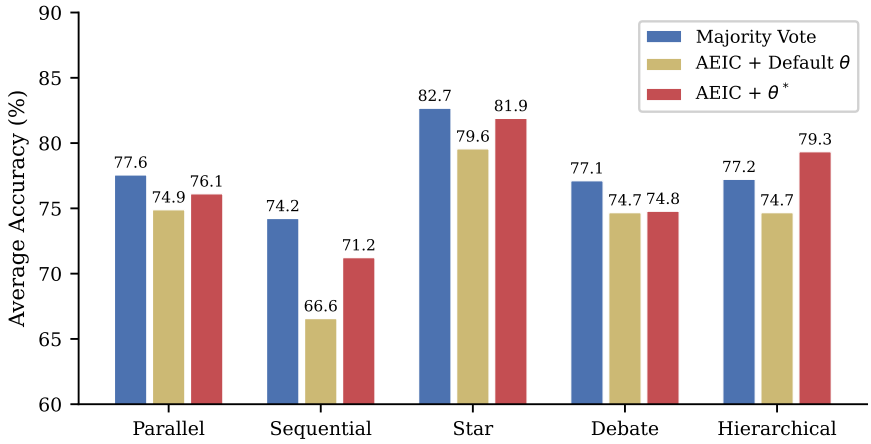


Fig. 2. Per-topology comparison averaged over three datasets. AEIC+θ^* only outperforms majority voting on hierarchical topology.

8 Discussion

Why Stackelberg over Nash? Yi et al. [21] model multi-agent debate as a simultaneous-move Bayesian game. Our Stackelberg formulation is more natural for mechanism design: the aggregation rule is *committed to* before agents act, enabling the leader to shape follower behavior through credible commitment—a property absent in Nash equilibrium where all players move simultaneously.

Quality Signals vs. Correctness. Our most significant finding is that AEIC quality signals measure reasoning process quality, not answer correctness. Majority voting outperforms AEIC-weighted voting on 4 of 5 topologies because a coherent, well-evidenced wrong answer scores as highly as a correct one. This implies that quality-based weighting cannot substitute for diversity-based aggregation when agents share similar knowledge and biases. The exception—hierarchical topology—suggests that quality weighting becomes valuable when communication structure creates information asymmetry.

AEIC as Feature Extraction. Despite limited value as direct voting weights, AEIC signals are highly informative features for topology routing. The 42-dimensional per-topology feature vector captures interactions between reasoning quality and communication structure that no single scalar signal can represent. The router achieves 92.4% using these features, suggesting that the appropriate level for quality assessment in multi-agent systems is *meta-level topology selection* rather than agent-level vote weighting.

Anti-Sycophancy from First Principles. Optimal mechanisms learn near-zero EV weights on star and debate topologies, providing a principled anti-sycophancy

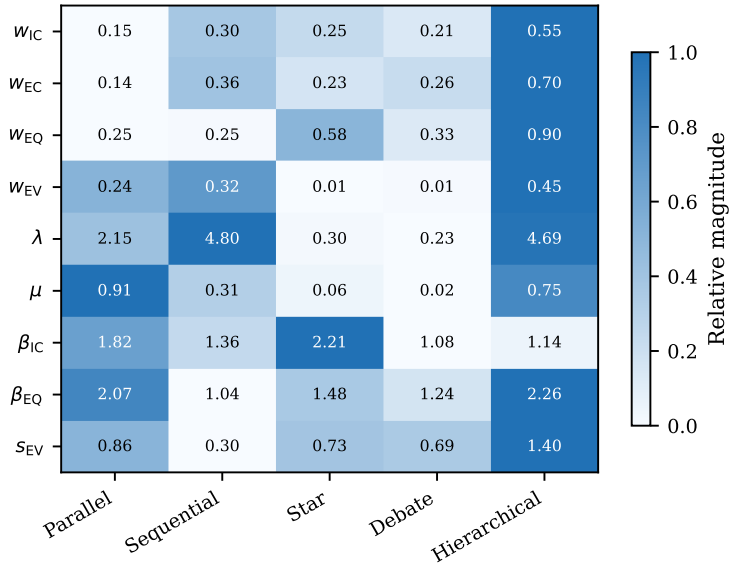


Fig. 3. Optimized θ^* parameters across five topologies (row-normalized). Star and debate learn near-zero w_{EV} (anti-sycophancy). Hierarchical learns high λ and elevated signal weights.

mechanism. On information-sharing topologies, high EV reflects social influence rather than independent verification. This emerges from accuracy optimization without explicit constraints, complementing heuristic approaches [12] with a game-theoretic foundation.

Oracle Gap Analysis. The 3.5pp gap between routing (92.4%) and oracle (95.9%) arises from questions where no topology produces correct consensus (∼4.1%) and cases where the top-2 picks both fail. Richer topology spaces or heterogeneous agents could address this.

Limitations. (1) The router requires running all five topologies for feature extraction (5× cost); online deployment needs topology prediction from question features alone. (2) All agents use the same LLM (DeepSeek-V3); heterogeneous pools may behave differently. (3) Evaluation is offline; online adaptation remains future work. (4) Top-2 selection uses ground truth; deployment requires learned selection or sequential evaluation.

9 Conclusion

We presented an adaptive game-theoretic framework for multi-agent LLM consensus that combines AEIC reasoning decomposition, Stackelberg incentive mechanisms, and neural topology routing. On 900 questions across three reasoning

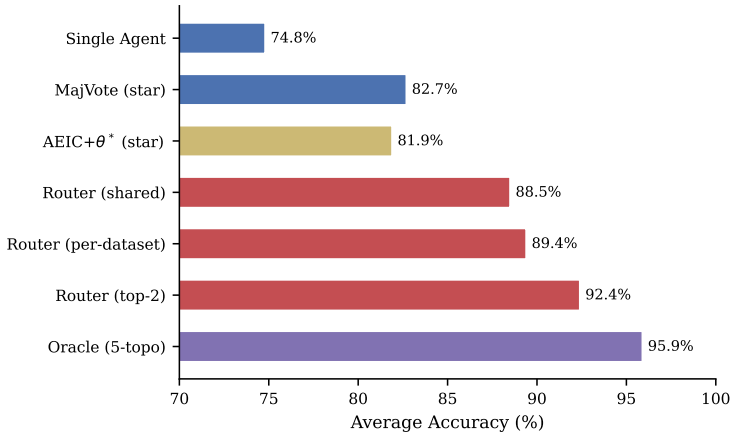


Fig. 4. Accuracy progression from single-agent to oracle. Top-2 router closes 74.8% of the gap between best fixed topology and oracle.

benchmarks with 5 agents and 5 communication topologies, our adaptive router achieves 92.4% consensus accuracy—17.6pp above single-agent performance and 9.7pp above the best fixed-topology majority vote—closing 74.8% of the gap to the oracle ceiling (95.9%).

Our key findings include: (1) AEIC quality signals measure reasoning process quality rather than answer correctness, making them more valuable as topology routing features than as direct voting weights; (2) per-topology Stackelberg optimization discovers interpretable parameter profiles, including anti-sycophancy effects where cross-agent consensus signals are suppressed on information-sharing topologies; and (3) a neural router with per-dataset scoring heads and top-2 selection substantially outperforms any fixed topology choice.

Future work includes developing online routing that selects topologies from question features without running all candidates, exploring heterogeneous agent pools with diverse LLM backends, and investigating richer topology spaces to further close the remaining 3.5pp oracle gap.

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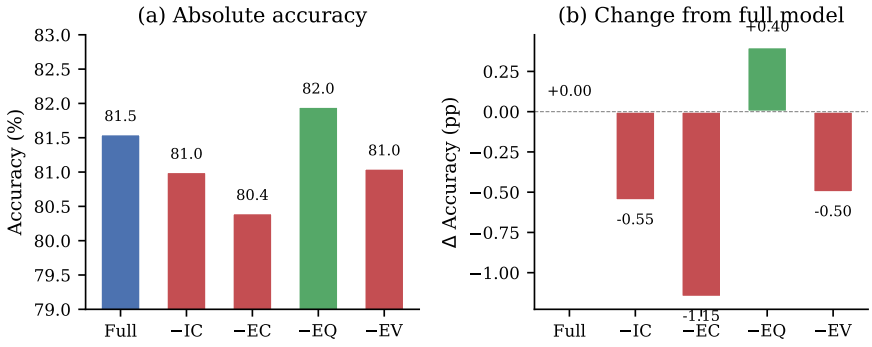


Fig. 5. Signal ablation under star topology. EC is the most critical signal (−1.15pp). Removing EQ slightly improves accuracy, consistent with its low optimized weight.

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